

Questions with Answer Keys

MathonGo

Q1: 24 Feb (Shift 1) - Numerical

Number of amphoteric compounds among the following is

- (A) BeO (B) BaO (C) $\text{Be}(\text{OH})_2$ (D) $\text{Sr}(\text{OH})_2$

Q2: 24 Feb (Shift 2) - Single Correct

Match List-I with List-II

List- I

List-II

(Salt)

(Flame colour wavelength)

(a) LiCl

(i) 455.5 nm

(b) NaCl

(ii) 970.8 nm

(c) RbCl

(iii) 780.0 nm

(d) CsCl

(iv) 589.2 nm

Choose the correct answer from the options given below :

(1) (a) – (ii), (b) – (i), (c) – (iv), (d) – (iii)

(2) (a) – (ii), (b) – (iv), (c) – (iii), (d) – (i)

(3) (a) – (iv), (b) – (ii), (c) – (iii), (d) – (i)

(4) (a) – (i), (b) – (iv), (c) – (ii), (d) – (iii)

Q3: 24 Feb (Shift 2) - Single Correct

The Correct set from the following in which both pairs are in correct order of melting point is

(1) $\text{LiF} > \text{LiCl}$; $\text{NaCl} > \text{MgO}$

(2) $\text{LiF} > \text{LiCl}$; $\text{MgO} > \text{NaCl}$

(3) $\text{LiCl} > \text{LiF}$; $\text{NaCl} > \text{MgO}$

(4) $\text{LiCl} > \text{LiF}$; $\text{MgO} > \text{NaCl}$

Q4: 25 Feb (Shift 2) - Numerical

Questions with Answer Keys

MathonGo

Among the following, number of metal/s which can be used as electrodes in the photoelectric cell is _____

(Integer answer).

(A) Li

(B) Na

(C) Rb

(D) Cs

Answer Key**Q1 (2)****Q2 (2)****Q3 (2)****Q4 (1)**

Q1 (2)

BeO and Be(OH)₂ are amphoteric in nature.

Q2 (2)

Range of visible region :

390nm – 760nm

VIBGYOR

Violet Red

LiCl Crimson Red

NaCl Golden yellow

RbCl Violet

CsCl Blue

So LiCl Which is crimson have wave length closed to red in the spectrum of visible region which is as per given data is.

Q3 (2)

Generally

$$\text{M.P.} \propto \text{Lattice energy} = \frac{KQ_1Q_2}{r^+ + r^-}$$

$\propto$ (packing efficiency)

Q4 (1)

Cs is used in photoelectric cell due to its very low ionization potential.